

Yagmur Gulec-CV (English)

Full Stack AI Developer | Intelligent Document Processing

Sherbrooke, Québec, Canada

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Full-stack AI developer with experience building and deploying intelligent document processing systems in a non-profit environment. Specialized in machine vision and OCR-based table extraction from PDF files, with a strong focus on AI quality assurance through hallucination measurement, temperature tuning, and experiment tracking. Comfortable working across the entire development lifecycle from rapid prototyping to production deployment.

TECH STACK

Frontend: React

Backend: FastAPI (Python)

AI/ML: OCR (Optical Character Recognition) for PDF table extraction, Large Language Models, hallucination detection, temperature parameter optimization

LLM Observability: hallucination detection, temperature tuning, prompt engineering, MLflow experiment tracking, OpenSearch validation

Deployment: Railway

Other: REST APIs, PDF processing, table extraction pipelines

EXPERIENCE

Observatoire Global du Saint-Laurent - *Programmer-analyst*

OCT 2025 - Present

- Develop full-stack AI applications using FastAPI (backend) and React (frontend)
- Achieved 90%+ hallucination-free LLM outputs in production through temperature tuning and observability pipelines
- Measure and mitigate AI hallucinations using text search and temperature parameter adjustment
- Track and debug AI model quality using MLflow (experiment tracking, performance monitoring)
- Deploy applications and rapid prototypes using Railway
- Integrate AI models into complete, user-facing production systems via REST APIs

M2M, Remote - *Machine Learning Developer Intern*

MAY 2025 - JUNE 2025

- Enhanced YOLO-based wildfire detection with domain-specific data and augmentation pipelines.
- Applied computer vision techniques for image preprocessing and labeling automation.

Riipen Beyond the Cloud and Level UP, Remote - *Software Developer*

FEBRUARY 2024 - MARCH 2025

- Developed a dashboard for prediction market data and deployed **for the customer's (AWS Lambda, SAM, Python, Dash)**.
- Automated provisioning Jenkins Server with **Terraform** using **Nginx** on **AWS EC2**.

Feedbacks and Projects: [Riipen Profile](#)

Universite de Sherbrooke, Québec - *Research Assistant in Mechanical Engineering*

FEB 2019 - SEP 2021

- Developed numerical models to simulate a vapor bubble in a boiling liquid pool (**Object oriented C++**)
- Deployed and automated the deployment large scale simulations on HPC cluster on Calcul Quebec (**Bash, Linux**)
- Implemented a Java plugin to process images to measure bubble sizes (**Java, ImageJ**)

EDUCATION

Bishop's University , Sherbrooke, QC - *Master's in Computer Science*

JAN 2022 - MAY 2024

GPA: 3.9/4.0 | Courses: Data Mining, Database Design, Deep Learning